

Load Profiling and Its Applications in Power Market

R. F. Chang and C. N. Lu, *Senior Member, IEEE*

Abstract-- In a power retail market, the knowledge of hourly load shape of the customer is often required in order to determine the tariff schedule. At the present time, requiring a meter as a prerequisite for power customers to choose a power supplier is not yet achievable. For customer screening, this paper proposes a technique that uses customers' monthly energy usage data to perform load profile assignment. Two steps are involved in the proposed method, they are the fuzzy c-means (FCM) method used for load profiling and a decision tree based pattern recognition technique for customer class assignments. Test data are from actual measurements and customer information system (CIS) of Taiwan Power Company (TPC). Promising results on assigning existing 500 customers to their proper load profiles are illustrated.

Index Terms-- Retail Market, Load Profile, Fuzzy Clustering, Decision Tree.

I. INTRODUCTION

In many power markets, customers are able to choose their electricity suppliers. Installing a meter for each customer could cause high costs that may place barriers to allowing all customers access to retail choice. In many occasions, if no hourly interval meter is installed, load profiling will enable their participation in retail choice. Identifying characteristics for assigning a customer to a load profile should be readily determined, so that application of the profiling process over the population of customers would be manageable [1]-[3].

For different types of customers, various factors may have different effects on the daily load profiles. Load profiles of residential customers vary in their daily/monthly activity patterns. Load profiles of commercial businesses have many needs unique to commercial functions, and load profiles of industry are influenced by type of industry, capacities of equipment, working time of each equipment, etc..

Many influence factors are useful for customer load profile assignment, but experiences have shown that these influential data are difficult to gather and the validity of the obtained data are often questionable. This paper proposes a technique that uses customers' monthly energy usages recorded by the utilities for load profile class assignment. The data are considered with more certainty. In the proposed method a

fuzzy c-means (FCM) method is used for load profiling and a decision tree technique is used for customer class assignments.

II. LOAD PROFILING

To establish typical load profiles, selected customers load data are gathered by digital meters. In this study, the sample customers used for the load survey were randomly chosen from the CIS of TPC. The Neyman stratification sampling method was applied to select the customers for measurement. Meters were programmed to record the customer real and reactive power demand every 15 minutes. Outliers are removed from the measurement data set. Since different customers have different peak demands, a conversion of the demands in per unit was conducted to have homogeneous curves.

Low voltage electricity customers often have relatively similar patterns of demand. In the context of the 1998 UK market agreements, eight generic profiles (two for domestic, and 6 for non-domestic customers) were used to represent under 100 kW customers [1]. In this study, dominant load profiles of 9 contract types customers in Taiwan are studied.

FCM method [4], [5] is used to identify typical load profiles. It is a data clustering technique wherein each data point belongs to a cluster to some degree that is specified by a membership grade. It is based on minimizing the following objective function that represents the distance from any given data point to a cluster center weighted by that data point's membership grade.

$$J_m = \sum_{k=1}^n \sum_{i=1}^c \mu_{ik}^m \|x_k - v_i\|^2 \quad (1)$$

where

x_k = The averaged load profile of k th customer, which is an R^{24} vector.

v_i = The i th cluster center.

μ_{ik} = Membership function value of the k th customer load profile in the i th cluster.

n = Number of load profiles.

c = Number of clusters.

m = The 'fuzzification' parameter, in general, $m=2$.

In each iteration, the cluster center is updated by the following equation [4], [5]:

R. F. Chang is with the Department of Electrical Engineering, Kao Yuan Institute of Technology, Lu-Chu., Taiwan (e-mail: rfchang@cc.kyit.edu.tw).

C. N. Lu is with Department of Electrical Engineering, National Sun Yat-Sen University, Kaohsiung, Taiwan (e-mail: cnl@ee.nsysu.edu.tw).

$$v_i^{(r)} = \sum_{k=1}^n (\mu_{ik}^{(r)})^m x_k / \sum_{k=1}^n (\mu_{ik}^{(r)})^m \quad \text{for } 1 \leq i \leq c \quad (2)$$

and the membership grade is modified by (3).

$$\mu_{ik}^{(r+1)} = \left[\sum_{j=1}^c \left[(d_{ik}^{(r)})^2 / (d_{jk}^{(r)})^2 \right] \right]^{-1/(m-1)} \quad \text{for } i=1, 2, 3, \dots, c \quad (3)$$

$$\text{where } (d_{ik}^{(r)})^2 = \|x_k - v_i^{(r)}\|^2 \quad \text{for } 1 \leq i \leq c, 1 \leq k \leq n \quad (4)$$

$\|x_k - v_i\|^2$ is the norm derived from the inner product.

Given the average metered customer load profiles (x_k 's), the procedure used to determine the typical load profiles (v_i 's) is as follows:

Step 1. Choose the values of c and m , let $2 \leq c < n$, $1 \leq m < \infty$; and initialize $\mu^{(0)}$.

Step 2. Calculate c means vectors $v_i^{(r)}$ and $(d_{ik}^{(r)})^2$.

Step 3. Update $\mu_{ik}^{(r+1)}$, for all i, k . Note that $\sum_{i=1}^c \mu_{ik}^{(r+1)} = 1$, $\mu_{ik}^{(r+1)} \in [0, 1]$.

Step 4. Stop if $|J_m^{(r+1)} - J_m^{(r)}| < \epsilon$, otherwise $r=r+1$, go to step 2.

Starts with an initial guess for the cluster centers, the procedure assigns every data point a membership grade for each cluster. By iteratively updating the cluster centers and the membership grades for each data point, it moves the cluster centers to the "right" location within a data set. The final cluster centers are chosen as typical load profiles of the samples.

III. DECISION TREE FOR LOAD PROFILE ASSIGNMENT

While household size, lifestyle and property type could be used as basic inputs for low voltage customer load profile class assignment, these data would require a great deal of effort to collect, and there is a doubt about the validity of these data. Customer billing data are considered more reliable than any data that can be collected by survey techniques. In this study customers' actual electricity usage data in CIS are utilized for existing customers load profile assignment. Many combinations of customer monthly energy usages and changes data including moving averages and others, can be used as inputs for load profile assignment study, the following data are used in this study:

$$E_{ij}^1 = \frac{kWh_i}{kWh_j}, i \neq j, i = \text{January} \sim \text{December}, j = \text{January} \quad (5)$$

$$E_{ij}^2 = \frac{kWh_i - kWh_{i+1}}{kWh_j}, i = \text{January} \sim \text{December}, j = \text{January} \quad (6)$$

$$E_{kl}^3 = \frac{kWh_k}{kWh_l}, k = \text{August} \sim \text{December}, l = \text{February} \sim \text{June} \quad (7)$$

$$E_{sj}^4 = \frac{kWh_s}{kWh_j}, s = \text{Spring} \sim \text{Winter}, j = \text{January} \quad (8)$$

where i and j are month indices, kWh_i is the kWh values in month i . E_{ij}^1 's, E_{kl}^3 's and E_{ij}^2 's represent the normalized monthly energy usages, monthly and seasonal changes respectively. E_{sj}^4 's represent the normalized averaged seasonal energy usages.

A decision tree (C4.5) method [6], [7] is used for customer class recognition. The network is trained based on a representative set of input/target pairs. Normalized customer energy usage and changes data obtained from (5)-(8) and other formulations are used as inputs to the decision tree and the outputs are corresponding to the known classes.

The approach starts with a set of training data consisting of a known class. The goal is to develop a tree that would correctly classify further examples into one of these classes, using the attributes values of the known examples. The algorithm examines each attribute in turn and calculates a Gain-Ratio measure of how well the attribute discriminates between the classes (a "goodness of split" measure) [6]. The best attribute is chosen, and the data are partitioned into subsets according to the values of that attribute. This procedure is recursively applied to each subset until all the examples are correctly classified. The result of this procedure is a tree in which nodes represent attributes and branches represent possible attribute values or ranges of values. Terminal nodes (leaves) of the tree correspond to sets of examples, all of which in the same class [6].

The Gain-Ratio measure is calculated by (9) [6].

$$GR(A_i) = G(A_i) / IV(A_i) \quad (9)$$

where

$G(A_i)$ = information gain due to the partition of the set C by attribute A_i (whose values include $a_1, a_2, \dots, a_k$)

$$= E(C) - E(A_i)$$

$E(C)$ = entropy of a set of objects C

$$= - \sum_j p_j * \log_2 p_j$$

j = output class j

p_j = (# of objects in class j) / (total # of objects in the set C)

$E(A_i)$ = entropy of the attribute A_i

$$= \sum_k (n_k / n) * E(C_k)$$

C_k = disjoint set k which is a partition of C by the attribute A_i according to A_i 's values.

$E(C_k)$ = entropy of the set C_k

n = total number of objects in the set C .

n_k = total number of objects in the subset C_k .

$IV(A_i)$ = information value of attribute A_i

$$= - \sum_{m=1}^k p_m * \log_2 p_m$$

$p_m = (\# \text{ of objects in } C \text{ whose } A_i \text{ value is } a_m) / (\text{total } \# \text{ of objects in the set } C)$

In the intended applications, the input data of decision tree are continuous attributes. A modification of the common decision tree approach is required for accommodating the continuous data. The binary partition algorithm is used to handle the continuous attribute. The procedure is as follows [6], [7]:

1. Sort attribute values in ascending order $v_1, v_2, \dots, v_n$ (assume there exist n values for the continuous attribute A).
2. Perform exhaustive binary partition at each mid-point (i.e., $(v_i + v_{i+1})/2$).
3. For each mid-point p , calculate the entropy of the partial tree if A is partitioned at the mid-point (i.e., $A \leq p$ and $A > p$).
4. Select the best partition point (i.e., resulting in the maximum gain-ratio) for A .

Because of the way it was built, the decision tree built deals correctly with most of the records in the training set. In fact, in order to achieve this goal, it may become quite complex, with long and very uneven paths. In this study, for not scarifying too much in overall performance and to minimize the complexity, the pruning of a decision tree could be stopped by replacing a sub-tree with a leaf node. The replacement takes place if a decision tree establishes that the expected error rate of the sub-tree is greater than in the single leaf. For a given confidence level (CF), the upper limit $U_{CF}(E, N)$ on this probability can be found from the confidence limits for the binomial distribution. When N training cases are covered by a leaf node L , among them E cases are incorrect, the upper limit of the predicted error rate of L is expressed as $U_{CF}(E, N)$. Then, the predicted errors of L are $N * U_{CF}(E, N)$. For each non-leaf sub-tree, if a further pruning of this sub-tree and replacing with a leaf would lead to a lower predicted error rate, then prune the sub-tree accordingly.

In summary, the procedure of the decision tree algorithm (C4.5) is as follows [6]:

1. Start from the root node and assign the root node as the current node C .
2. If all objects in the current node C belong to the same class, then stop (the termination condition for the current C) else go to step3.
3. Calculate the entropy $E(C)$ for the node C .
4. Calculate the entropy $E(A_i)$ of the partial tree partitioned by an attribute A_i which has not been used as classing attributes of the node C .

5. Computer the gain-ratio for the partial tree.

6. Repeat step 4 and 5 for each attribute which has not been used as classifying attributes for the node C .

7. Select the attribute with the maximum gain-ratio as the classifying attribute for the node C .

8. Create child nodes $C_1, C_2, \dots, C_n$ for the node C ; and assign objects in the node C to appropriate child nodes according to the values of the classifying attribute.

9. For each node C_i , assign it as the current node and go step 2.

IV. TEST RESULTS AND DISCUSSIONS

Actual load profile data of 300 TPC low voltage customers and 200 high voltage customers are used in this study. Based on the FCM method, Figs. 1, and 2 show the obtained load profiles of low voltage customers and high voltage customers. When the 300 low voltage customers are classified into 6 different load profiles, Fig. 3 shows the averaged load profiles of low voltage customers that are assigned to class number 1.

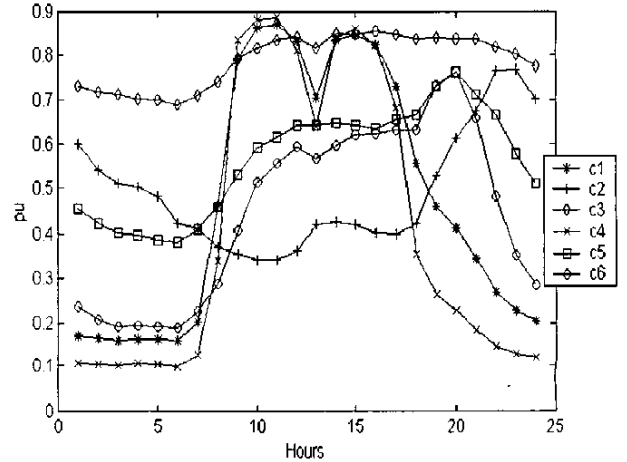


Fig. 1. 6-Class load profiles of low voltage customers.

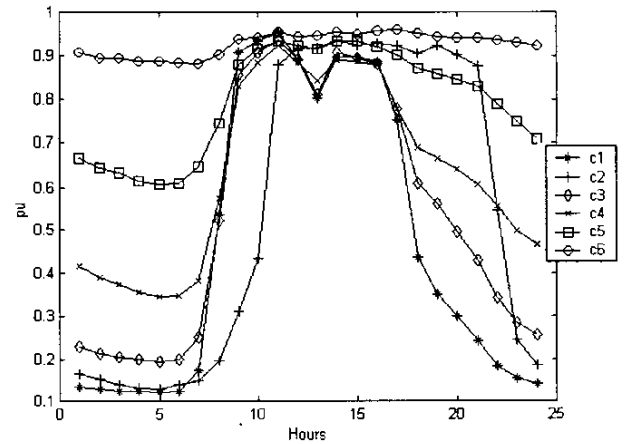


Fig. 2. 6-Class load profiles of high voltage customers.

Using the measured load profiles and FCM method customers in the load survey samples are assigned to specific classes. These data and the customer monthly energy usage

data are then used to test the proposed load profile assignment technique. Trees for low voltage customers and high voltage customers are built separately.

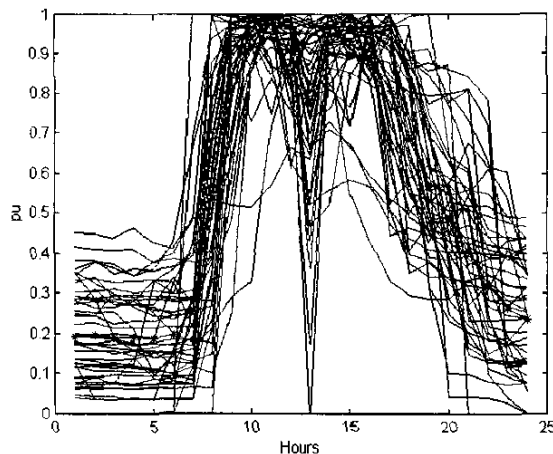


Fig. 3. Low voltage Class 1 customers load profiles.

We used 10 sets of randomly sampled training data each has 240 and 160 customers drawn proportionally from each class to induce the tree, and we used the remaining 60 and 40 customers data to test the performance of the tree for low voltage customers and high voltage customers respectively. 31 continuous attributes obtained from (5) ~ (9), and 1 discrete attribute representing the contract type of customers, and a known profile class are used to induce the decision trees. The pruning CF is set at 25%.

Figs. 4 and 5 show part of the decision tree for the low voltage customers and high voltage customers respectively. We can use the essential attributes of the decision tree to obtain the load profile classes of the customers. Table I show the essential attributes.

Tables II and III show 10 tests results of low voltage and high voltage customers respectively. As shown in Tables II and III, on average, only 11.8 and 5.2 customers in the training sets and 11.6 and 5.1 customers in the testing sets are misclassified.

For different types of customers, various attributes may have different effects on the daily load profiles. In order to improve the accuracy of the load profiles assignment, adding other impact attributes to induce the decision tree may be a useful way.

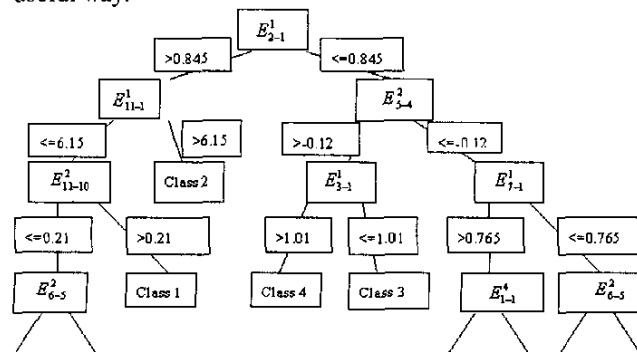


Fig. 4. Decision tree of low voltage customers.

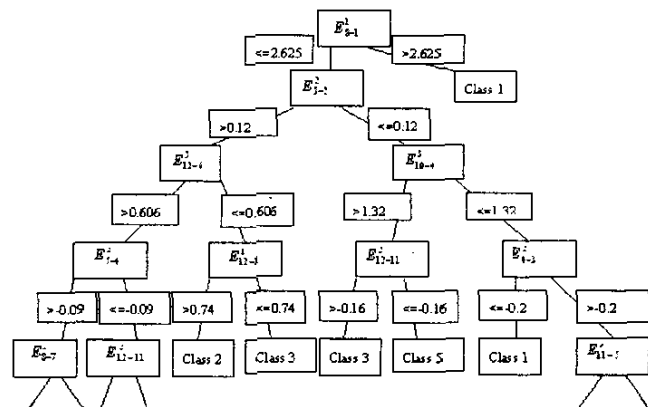


Fig. 5. Decision tree of high voltage customers.

TABLE I
THE ESSENTIAL ATTRIBUTES

Original attributes	Essential attributes	
	Low voltage	High voltage
31 continuous attributes obtained from (5)~(8) and 1 discrete attribute	$E_{2-1}^1, E_{3-1}^1, E_{4-1}^1,$ $E_{5-1}^1, E_{6-1}^1, E_{7-1}^1,$ $E_{11-1}^1, E_{12-1}^1, E_{3-2}^2,$ $E_{5-4}^2, E_{6-5}^2, E_{7-6}^2,$ $E_{8-7}^2, E_{9-8}^2, E_{10-9}^2,$ $E_{11-10}^2, E_{12-11}^2, E_{8-2}^3,$ E_{11-5}^3, E_{4-1}^4	$E_{2-1}^1, E_{7-1}^1, E_{8-1}^1,$ $E_{9-1}^1, E_{10-1}^1, E_{11-1}^1,$ $E_{12-1}^1, E_{3-2}^2, E_{4-3}^2,$ $E_{5-4}^2, E_{6-7}^2, E_{10-9}^2,$ $E_{11-10}^2, E_{12-11}^2, E_{10-4}^3,$ $E_{8-2}^3, E_{11-5}^3, E_{12-6}^3,$ $E_{1-1}^4, E_{3-1}^4, E_{4-1}^4$

TABLE II
LOAD PROFILE ASSIGNMENTS TEST RESULTS (LOW VOLTAGE CUSTOMERS)

Test No.	Training accuracy	Test accuracy
1	221/240	49/60
2	228/240	52/60
3	230/240	46/60
4	229/240	50/60
5	228/240	46/60
6	228/240	47/60
7	229/240	48/60
8	227/240	49/60
9	231/240	46/60
10	230/240	51/60
Average	95%	80.7%

TABLE III
LOAD PROFILE ASSIGNMENTS TEST RESULTS (HIGH VOLTAGE CUSTOMERS)

Test No.	Training accuracy	Test accuracy
1	158/160	35/40
2	154/160	32/40
3	155/160	35/40
4	154/160	36/40
5	154/160	37/40
6	155/160	36/40
7	154/160	35/40
8	155/160	35/40
9	156/160	34/40
10	153/160	34/40
Average	96.8%	86.7%

V. CONCLUSION

Billing activities in the retail and wholesale markets require the knowledge of the hourly load shapes by customer classes. Instead of installing hourly meter for each customer participating in the retail market, this paper uses a load profiling technique that exploits customer energy usage data for customer class assignment. Test results indicate that the decision tree method are capable of assigning customer load profiles, or at least, performing a preliminary class screenings. Test results also show that different combinations of user monthly energy data may be useful for load profile assignments in the retail power market.

VI. REFERENCES

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VII. BIOGRAPHIES

Rung-Fang Chang received his B.S. degree in electrical engineering from the Chung Yuan Christian University, Chung-Li, Taiwan in 1990 and received the M.S. degree and the Ph.D. degree from the National Sun Yat-Sen University, Kaohsiung, Taiwan in 1992 and 2002 respectively. He is an associate professor of the Department of the Electrical Engineering, Kao Yuan Institute of Technology, Lu-Chu, Taiwan. His research interests are optimization of power system and load research.

Chan-Nan Lu (S'83-M'85-SM'92) received the B.S. degree from the National Taiwan University, Taipei, Taiwan, in 1981, the M.S. degree from Rensselaer Polytechnic Institute, Troy, NY, in 1983, and the Ph.D. degree from Purdue University, West Lafayette, IN, in 1987. He held positions at General Electric Co., Pittsfield, MA, and Harris Corporation Control Division, Melbourne, FL. He has been with National Sun Yat-Sen University since 1989. His current research interests are power quality, power system deregulation, and distribution automation.